1. $\cap$ vs. $\cup$.

and or.

$A \cap B$. A and B happen.

$A \cup B$ A or B happen.

2. m.e. vs. independent.

If A and B are m.e., then A and B are <u>NOT</u> independent.

If A and B are independent, then A and B are <u>NOT</u> m.e.

3. <u>Additive Rule.</u>

1> General additive rule.

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

One special case: for A and B are <u>m.e.</u>

then $P(A \cup B) = P(A) + P(B)$

4. Product Rule.

1> General product rule.

$$P(A \cap B) = P(A|B)\,P(B)$$
$$= P(B|A)\,P(A)$$

One special case: for A and B are <u>independent</u>.

then $P(A \cap B) = P(A) \cdot P(B)$

5. Complement Rule.

$$P(A) = 1 - P(A')$$

$$P(A') = 1 - P(A)$$

with conditions, we have

$$P(A \mid B) = 1 - P(A' \mid B)$$

$$P(A' \mid B) = 1 - P(A \mid B)$$

Continue the previous example.

a) Find probability of a false "+" test

$$P(+ \mid P') = 1 - P(+' \mid P')$$

$$= 1 - P(- \mid P')$$

$$= 1 - 0.94 = 0.06$$

b) Find probability of a false "−" test

$$P(- \mid P) = 1 - P(-' \mid P)$$

$$= 1 - P(+ \mid P)$$

$$= 1 - 0.91 = 0.09$$

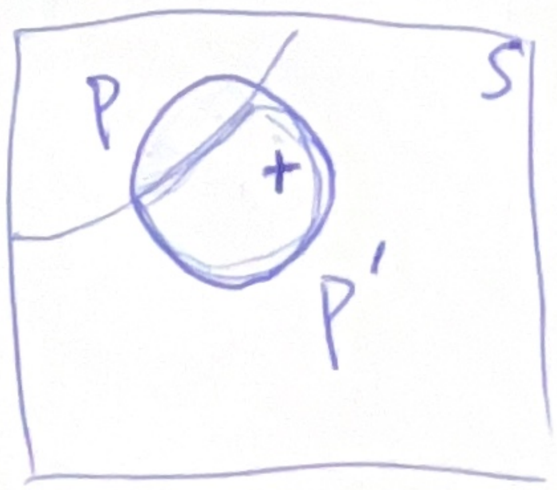
c) Find the probability a R.S. sample, tested "+"

$$P(+) = P(+ \cap P) + P(+ \cap P')$$

$$= P(P)P(+ \mid P) + P(P')P(+ \mid P')$$

$$= 0.13 \times 0.91 + (1 - 0.13) \times 0.06$$

$$= 0.1705$$



d) $P(P|+) = \dfrac{P(P \cap +)}{P(+)} = \dfrac{P(P)\,P(+|P)}{P(+)}$

$= \dfrac{0.13 \times 0.91}{0.1705} = 0.6938.$

e) $P(P'|-) = \dfrac{P(P' \cap -)}{P(-)} = \dfrac{P(P')\,P(-|P')}{1 - P(+)}$

$= \dfrac{(1-0.13)(0.94)}{1 - 0.1705} = 0.9859.$